

Predicting Household Appliance Energy Consumption

Milestone 2: Dataset and Problem Approval

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1. Problem and purpose

We want to compare how well different machine-learning models can estimate appliance energy use in a household. The inputs are temperature, humidity, weather, lighting, and time information. The output is appliance energy consumption in watt-hours (Wh) for the same 10-minute period. This is supervised regression, meaning the model learns from examples with a known numeric output.

The intended users could include homeowners and building-energy analysts. For this course, the project is a proof of concept. We want to understand whether these measurements provide useful predictive information and whether more flexible models improve on simpler ones. The project estimates same-period consumption; it does not claim to forecast energy use before the input measurements are available.

2. Source, license, and prior work

We use the UCI Appliances Energy Prediction dataset, donated by Luis Candanedo in 2017. Dataset DOI: 10.24432/C5VC8G. Source:

<https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction>. The license is Creative Commons Attribution 4.0 (CC BY 4.0), which permits reuse with appropriate attribution. We will keep the source, license link, and a description of our changes in the repository.

The associated paper is Candanedo, Feldheim, and Deramaix (2017), Data driven prediction models of energy use of appliances in a low-energy house, Energy and Buildings, 140, 81-97, DOI: 10.1016/j.enbuild.2017.01.083. Our contribution is a controlled comparison using concepts taught in the course, not a new algorithm or a claimed reproduction of the paper's exact results.

3. Main research question

How does model complexity affect estimates of household appliance energy consumption when the models use the same data and evaluation procedure? We will also check how much time features and lighting contribute, and which observations produce the largest errors.

4. Data card and sample input/output

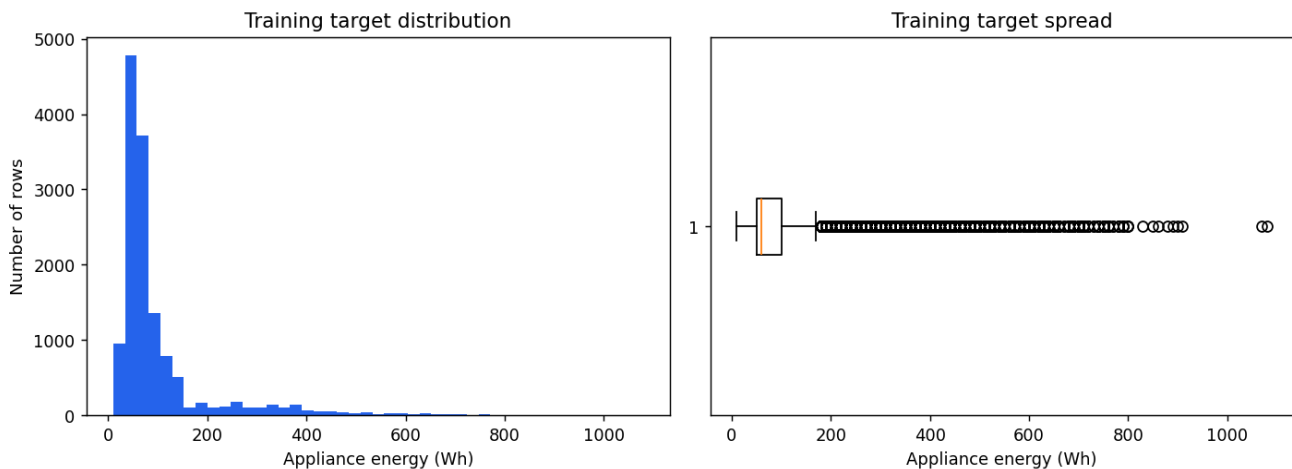
Item	Details
One observation	One 10-minute period in one low-energy house in Belgium.
Size	19,735 rows; 29 columns including the target, or 28 potential input columns.
Date coverage	January 11, 2016 at 17:00 to May 27, 2016 at 18:00.
Target	Appliances: appliance energy use in Wh.
Feature groups	Temperature (C), humidity (%), lighting (Wh), weather, and timestamp. Weather includes pressure (mm Hg), wind speed (m/s), visibility (km), and dew point (C).
Sensor locations	Kitchen, living room, laundry, office, bathroom, outdoors on the north side, ironing room, teenager room, and parents room. Separate weather-station readings are also present.
Random controls	rv1 and rv2 are identical in this copy and are excluded from the main models.
Data quality	No missing values, duplicate rows, or duplicate timestamps. Every adjacent timestamp is 10 minutes apart.
Known limits	One house, about 4.5 months, no direct occupancy information, and occasional high energy-use values.

For the first observation, example inputs include kitchen temperature 19.89 C, kitchen humidity 47.60%, living-room temperature 19.20 C, lighting energy 30 Wh, and timestamp January 11, 2016 at 17:00. The observed output is 60 Wh. The notebook shows complete model input rows and their matching labels; these are examples from the data, not predictions.

The timestamp becomes hour and day-of-week features. Appliances is always excluded from the input matrix. After removing the raw timestamp and both random controls and adding two time features, the initial model has 27 predictors. Lighting is included only for same-period estimation where that measurement is available. A separate comparison will remove lighting.

5. Initial exploratory analysis

We checked the full file for structural issues, then limited target and relationship analysis to the training block. The training target has mean 98.78 Wh, median 60 Wh, and range 10-1080 Wh. Most observations have lower consumption, with fewer large spikes. This uneven distribution makes it extremely important to inspect errors on high-consumption observations.



The histogram has a long right-hand tail: a smaller number of high values extend the range of the histogram. We will keep these observations unless there is evidence of a measurement error. Removing genuine spikes could make the results look better while making the task less realistic.

Training averages vary by hour and day. The highest average hour is 18:00 and the lowest is 03:00. This supports testing time features, but does not establish a pattern for other houses or later seasons.

The notebook includes numeric summaries, physical-range checks, daily and hourly plots, and correlations with the target. No training humidity values fall outside 0-100%, and no negative energy or non-finite numeric values were found. A plausible range alone does not prove that every sensor value is accurate.

The strongest absolute linear correlations among the proposed training inputs are hour (0.232), lights_energy_wh (0.226), temp_living_room_c (0.150). Correlation describes how values move together; it does not prove cause and effect or rule out nonlinear relationships.

6. Preparation plan

We will parse and sort timestamps; keep the target separate; remove rv1 and rv2; create hour and day-of-week inputs. Use scaling for kNN and fit all learned preprocessing within training folds. No target lags or rolling features are currently planned. Any later additions will need a new leakage check and a documented time gap where appropriate.

7. Training, validation, and test strategy

We chose a chronological 70/15/15 split. The earlier observations train the models, the next block validates decisions, and the final block is reserved for final testing. This avoids randomly mixing nearby observations across the split.

Split	Rows	Start	End
train	13814	2016-01-11 17:00:00	2016-04-16 15:10:00
validation	2960	2016-04-16 15:20:00	2016-05-07 04:30:00
test	2961	2016-05-07 04:40:00	2016-05-27 18:00:00

Hyperparameter tuning will use five expanding-window time-series folds inside the training block. We will use the validation block to confirm the selected configuration, then freeze the final choice. We may refit on training plus validation before evaluating the test block once. Each stage fits scaling and benchmark constants only on its permitted training data. Randomized models use seed 42 and the observations are not going to be shuffled.

8. Models, metric, and success criterion

We will compare training-mean and training-median constant benchmarks, linear regression, kNN regression, decision-tree regression, and random-forest regression. These methods cover linear, distance-based, and tree-based learning. The median benchmark is included because it minimizes absolute error among constant predictions.

The primary metric is Mean Absolute Error (MAE), fixed before model selection. It gives the average absolute prediction error in Wh. For example, MAE of 10 Wh means predictions differ from observed values by 10 Wh on average. RMSE and R-squared will be supporting metrics. RMSE gives more weight to large errors.

Our empirical success criterion is lower final-test MAE than both constant benchmarks. We will report percentage improvement, validation-fold variability, and errors during high-consumption periods. We do not guarantee a particular improvement in advance. A model that fails to beat the benchmarks is still a result that we will explain.

9. Planned additional analysis

Compare environmental/weather features, time features, and their combination. Also compare models with and without lighting. Use the same split and tuning rules for these comparisons. This will show whether extra measurements help, rather than only reporting which model wins.

10. Risks and limitations

The data comes from one house and part of one year, so results cannot establish performance across households or all seasons. Appliance use depends on activity and occupancy that are not being directly measured. High-consumption periods may be difficult to predict. Temporal dependence can make random-split results overly favourable, which is why we chose chronological evaluation.

Same-period temperature, lighting, and weather must actually be available for the claimed use. A live application would need to check weather-data timing and availability. Future forecasting would require a different input plan. Predictive relationships do not show that changing a temperature or humidity value causes a change in appliance energy use.

11. Feasibility and compute plan

The CSV is about 12 MB and contains fewer than 20,000 rows. We plan to use an ordinary CPU laptop with at least 8 GB RAM and Python, pandas, numpy, matplotlib, scikit-learn, and Jupyter. No GPU or paid cloud service is planned to be used. Actual team hardware will be recorded before model training.

Initial tuning limits: kNN neighbours in {5, 15, 30} with uniform or distance weights; decision tree depth in {5, 10, None} with minimum leaf size in {5, 20}; random forest with 100 trees and the same depth/leaf choices. Each tuned model starts with six configurations over five time-series folds. Linear regression has no tuning grid. We will explain model-specific tuning differences, record runtime, and document any reduced search before looking at the test results.

12. AI assistance and team review

AI helped with brainstorming ideas, organizing the notebook, answering code-related questions, and clarifying scientific terminology and phrasing. The team remained responsible for the final wording and code. No AI-generated code was copied and pasted into the project.

Repository

<https://github.com/AndyLopezCoding/MachineLearningProject>